

Channel Prediction for Mobile Distributed MIMO (MD-MIMO) Through Kalman-Structured Online Learning

Usama Saeed, Yibin Liang, Nima Mohammadi, Daniel J. Jakubisin, and Lingjia Liu

Abstract—Mobile distributed MIMO (MD-MIMO) extends conventional distributed MIMO by enabling mobility of the distributed radio units (RUs) and replacing wired fronthaul with wireless links. However, MD-MIMO performance depends heavily on the availability of accurate channel state information (CSI), which becomes stale quickly in high-mobility environments. Channel prediction, therefore, emerges as a key enabling technology for realizing the potential gains of MD-MIMO. Kalman filter-based prediction methods achieve minimum mean-square error (MMSE) performance assuming the availability of accurate channel dynamics, but their performance degrades substantially under model mismatch in practical environments. In this paper, we address this limitation by designing a channel prediction framework that combines model-based structure with data-driven AI/ML learning strategies. In this physics-based AI approach, we leverage statistics of the Kalman predictor to configure the dynamics of a Windowed Echo State Network (WESN), embedding model-based autoregressive structure into the neural network while retaining low-complexity online training for robustness to imperfect system knowledge. Through extensive simulations under practical 3GPP evaluation methodologies, we demonstrate that the introduced approach outperforms both Kalman filtering and state-of-the-art learning-based baselines in terms of channel prediction accuracy and downlink throughput for MD-MIMO under realistic mobility and feedback constraints.

I. INTRODUCTION

Distributed Multiple-Input Multiple-Output (D-MIMO) has emerged as a key architectural paradigm for next-generation wireless networks, enabling geographically separated radio units (RUs) to cooperate for joint transmission and reception [1], [2]. By spatially distributing transmit antennas across the coverage area, D-MIMO enhances coverage and spatial multiplexing performance compared to conventional co-located MIMO systems. Such a distributed framework is particularly well-suited for NextG networks, where dense deployments, heterogeneous infrastructures, and the need for flexible network reconfiguration are expected to be fundamental.

Mobile Distributed MIMO (MD-MIMO) further extends this paradigm by replacing conventional fiber-based fronthaul (FH) with wireless FH and by allowing Distributed Units

(DUs) and Radio Units (RUs) to be mobile, thereby significantly enhancing deployment flexibility and adaptability for NextG networks [3]–[5]. Compared to conventional D-MIMO, MD-MIMO promises increased deployment flexibility, rapid reconfigurability, and reduced operating and deployment costs, making it particularly attractive for highly dynamic environments. MD-MIMO can enable rapid deployment in scenarios such as Vehicle-to-Everything (V2X), UAV-assisted networks, emergency response, and tactical communications. However, these advantages come at the cost of increased system complexity. MD-MIMO must coordinate joint transmission across spatially distributed mobile RUs while operating under limited and delayed CSI. As a result, the same features that make MD-MIMO attractive also impose stricter physical-layer coordination challenges.

Accurate CSI is essential for Coherent Joint Transmission (CJT), as the distributed RUs must jointly construct a downlink precoder whose phases and amplitudes align at the intended User Equipments (UEs) while suppressing inter-user interference. In practical frequency-division duplex (FDD) systems, CSI is estimated at the UE and fed back to the DU through quantized codebook-based mechanisms, resulting in imperfect and delayed channel knowledge at the transmitter [5]. This leads to the well-known phenomenon of CSI aging, where a mismatch arises between the channel used for precoding and the actual channel during transmission [6]. The impact of this mismatch is particularly severe in D-MIMO systems [7], [8] and even worse in MD-MIMO [9], as both UE mobility and RU mobility contribute to rapid channel variation. Additionally, MD-MIMO introduces additional coordination and fronthaul operations before joint transmission. These operations increase the effective latency between CSI acquisition and data transmission. As a result, the precoder becomes misaligned with the true channel, degrading coherent combining gains and reducing the effectiveness of interference suppression. These effects are further exacerbated in high-mobility environments, where the channel may change significantly between estimation, feedback, and downlink transmission.

Channel prediction is therefore a key mechanism for mitigating CSI aging without increasing pilot and CSI feedback overhead. Rather than feeding back CSI corresponding to the current channel realization, the UE predicts the future channel that will be observed when the DU applies the downlink precoder. This prediction horizon is directly tied to the feedback and processing delay, and grows with the latency introduced by CSI feedback delay and MD-MIMO fronthaul coordination.

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As a result, channel prediction becomes particularly critical in MD-MIMO systems, where both UE and RU mobility accelerate channel variation and amplify CSI mismatch. In this context, prediction is an enabling technology for MD-MIMO CJT operation [5], [6].

Conventional channel prediction methods, including autoregressive (AR) modeling, Wiener filtering, and Kalman filtering, exploit temporal correlation in wireless channels. In particular, Kalman filtering provides a minimum mean-square error (MMSE) predictor under a linear Gaussian state-space model [6], [10]. However, the performance of Kalman filtering-based predictors relies on accurate knowledge of the underlying model parameters, including the state transition dynamics and noise statistics. In practice, these quantities are not known a priori and must be estimated from data [6]. This highlights a key practical limitation: model-based predictors depend on estimated dynamics that may deviate from the true channel evolution, particularly in realistic propagation environments with complex scattering, mobility, and non-stationarity. As a result, performance can degrade when the assumed model does not accurately capture the underlying channel behavior [11], [12].

To address these limitations, machine learning (ML)-based channel prediction methods have been introduced to learn channel dynamics directly from data. Recent approaches include Recurrent Neural Networks (RNNs) [11], attention-based models such as transformers [13], [14], and structured state-space sequence models [15]. Learning-based channel predictors can be broadly categorized into offline and online approaches. In offline learning, model parameters are trained in advance using a pre-collected dataset and are typically fixed during deployment. In online learning, the predictor updates its trainable parameters during operation using newly observed data. While offline data-driven methods can reduce the reliance on explicit system identification, they introduce new practical challenges. First, many ML-based predictors incur significantly higher computational complexity than Kalman filtering-based approaches, particularly during training [6]. Second, offline learning methods require large and representative training datasets to achieve good performance across diverse channel conditions [16]. In practice, collecting such datasets is challenging because the training data must capture a wide range of RU mobilities, UE mobilities, network geometries, blockage conditions, and propagation environments. Moreover, offline-trained models may not generalize well when deployed in environments that differ from the training data [16]. This issue is particularly acute for channel prediction, where channel statistics can evolve rapidly over time. These challenges highlight the need for channel prediction methods that support online learning, enabling adaptation to changing channel conditions with limited data [17].

Reservoir computing (RC), and in particular Echo State Networks (ESNs), provide an attractive framework for online learning in wireless problems [18], [19]. In ESNs, the recurrent dynamics are fixed and only a linear readout layer is trained, significantly reducing training complexity compared to fully trainable neural networks [20]. This structure enables efficient online adaptation using limited training data,

while still providing temporal memory through the recurrent reservoir. The Windowed ESN (WESN) [21] extends the standard ESN by augmenting the ESN input with a sliding window of recent observations, thereby exposing short-term channel history directly to the reservoir and readout. As a result, WESN balances modeling flexibility and computational efficiency, and has been applied in wireless communication tasks such as MIMO-OFDM symbol detection [18], [21]–[23] and even MIMO-OFDM channel prediction [4], [5]. In the context of MD-MIMO, this is especially appealing, as it allows online adaptation to time-varying channels, making RC-based predictors better aligned with practical PHY-layer constraints than fully offline deep learning approaches.

However, conventional RC approaches typically rely on randomly initialized reservoirs [5], so their recurrent dynamics are not explicitly designed to take advantage of domain knowledge. In this work, we address this limitation by configuring a WESN using the structure of a Kalman filtering-based predictor. Specifically, we design the reservoir using the steady-state Kalman predictor as a target dynamical system. This leads to a hybrid approach that combines model-based and data-driven learning: the configured reservoir embeds the structure of a Kalman-filtering-based linear predictor, while the readout layer is trained online to compensate for model mismatch and imperfect knowledge of channel dynamics. We integrate this predictor into an MD-MIMO CJT pipeline with practical 3GPP Type-II codebook-based CSI feedback and evaluate its impact on both channel prediction Normalized Mean Square Error (NMSE) and end-to-end downlink throughput.

The main contributions of this work are:

- We formulate a WESN-based online learning architecture for MD-MIMO OFDM channel prediction. We adapt the WESN formulation to the MD-MIMO OFDM setting by applying it on a per UE-RU link basis.
- We introduce a weight configuration approach for the WESN predictor, where the reservoir is designed using the structure of the steady-state Kalman predictor rather than initialized randomly. By interpreting the Kalman predictor as a linear dynamical system and approximating its transfer function through a bank of first-order infinite impulse response (IIR) filters, the resulting reservoir intrinsically captures the dominant temporal dynamics of the wireless channel.
- We demonstrate through link-level simulations of MD-MIMO CJT systems that the configured WESN achieves improved channel prediction accuracy compared to Kalman-filtering-based predictors and learning-based baselines. These gains are shown to translate directly into improved precoding quality and increased system throughput under realistic conditions, including CSI feedback delay and codebook-based CSI feedback.

Our results demonstrate that while increasing the number of RUs enhances multiplexing and reliability gains, these gains are highly sensitive to CSI aging. Channel prediction significantly mitigates this degradation, enabling robust coherent joint transmission under realistic feedback constraints.

The remainder of this paper is organized as follows: Section II introduces the MD-MIMO system model and the CJT

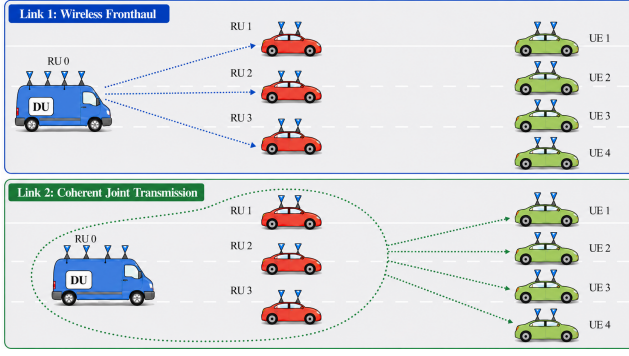


Fig. 1. Two-link MD-MIMO operation. Link 1 (top): RU 0 broadcasts the downlink payload over the wireless fronthaul to distributed RUs 1–3. Link 2 (bottom): RUs 0–3 jointly transmit to the UEs through coherent joint transmission.

downlink signal formulation. Section III describes practical CJT operation under imperfect CSI, including codebook-based feedback and CSI aging. Section IV reviews the preliminaries on Kalman filtering and ESNs. Section V presents the configured WESN framework for channel prediction, including the derivation of the reservoir dynamics from steady-state Kalman predictor statistics. Section VI provides a detailed complexity analysis comparing configured WESN with conventional Kalman filtering. Section VII integrates the introduced predictor into the CJT downlink pipeline. Section VIII presents simulation results demonstrating the performance gains in terms of both channel prediction accuracy and system throughput. Finally, Section IX concludes the paper.

II. SYSTEM MODEL

We consider an MD-MIMO downlink architecture in which a mobile Distributed Unit (DU) coordinates a set of G geographically distributed Radio Units (RUs), as illustrated in Fig. 1. RU g is equipped with N_g antennas, $g \in \{0, \dots, G-1\}$. The DU performs centralized baseband processing, including payload generation, CSI processing, and downlink precoder computation, while the RUs serve as distributed transmit front-ends. Through coordination across the RUs, the system forms a virtual transmit array for CJT in the downlink.

To enable wireless fronthaul, we assume that RU 0 is co-located with the DU and has a reliable local connection to it (e.g., wired or in-box interface). RU 0 acts as the wireless FH transmitter, while the remaining RUs $g \in \{1, \dots, G-1\}$ receive the fronthaul transmission and recover the payload intended for joint downlink transmission. End-to-end MD-MIMO downlink operation therefore proceeds over two links:

- **Link 1 (wireless FH):** RU 0 transmits the downlink payload to the distributed RUs $g = 1, \dots, G-1$,
- **Link 2 (CJT downlink):** all RUs jointly transmit to the UEs using coherent downlink precoding.

Both links operate under FDD, requiring explicit CSI feedback. The two links are separated in time through Time Division Multiplexing (TDM) to avoid mutual interference. Link 1 focuses on reliable wireless payload delivery from the DU-side RU to the distributed RUs, whereas Link 2 exploits the spatial degrees of freedom of the distributed array for

multi-user downlink transmission. In this work, we assume perfect Link 1 and focus on channel prediction as an enabling technology for Link 2.

A. Link-2 CJT Signal Model

After Link 1, all participating RUs are assumed to hold identical copies of the symbol vector corresponding to the downlink payload. Link 2 then performs coherent joint transmission from the distributed RU set to the UEs. The aggregate set of transmit antennas across all RUs forms a distributed virtual array with $N_{\text{tot}} = \sum_{g=0}^{G-1} N_g$ transmit antennas.

Let K denote the number of simultaneously served UEs. UE k is equipped with M_k antennas and is intended to receive d_k spatial streams. Define the aggregate transmit symbol vector:

$$\mathbf{s} = [\mathbf{s}_1^T \quad \mathbf{s}_2^T \quad \dots \quad \mathbf{s}_K^T]^T \in \mathbb{C}^{d \times 1}, \quad (1)$$

where $\mathbf{s}_k \in \mathbb{C}^{d_k \times 1}$ and $d = \sum_{k=1}^K d_k$.

Let $\mathbf{W}_{\text{DL}} \in \mathbb{C}^{N_{\text{tot}} \times d}$ denote the global downlink precoding matrix applied across all RUs. Let $\mathbf{H}_{k,g} \in \mathbb{C}^{M_k \times N_g}$ denote the channel from RU g to UE k . The aggregate channel from all RUs to UE k is given by

$$\mathbf{H}_k = [\mathbf{H}_{k,0} \quad \mathbf{H}_{k,1} \quad \dots \quad \mathbf{H}_{k,G-1}] \in \mathbb{C}^{M_k \times N_{\text{tot}}}. \quad (2)$$

The received signal at UE k is therefore

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{W}_{\text{DL}} \mathbf{s} + \mathbf{n}_k, \quad (3)$$

where $\mathbf{n}_k \sim \mathcal{CN}(\mathbf{0}, \sigma_k^2 \mathbf{I})$ denotes additive Gaussian noise.

When timing, carrier-frequency, and phase synchronization are maintained across the RUs, the distributed antennas can combine coherently at the UE. Under accurate CSI, this enables both distributed beamforming gain and spatial multiplexing gain. However, in practical FDD operation, the downlink precoder must be computed from fed-back and quantized CSI, and is therefore sensitive to feedback imperfections and delay.

III. PRACTICAL CJT OPERATION UNDER IMPERFECT CSI

In practical FDD systems, CJT relies on CSI acquired at the UE and fed back to the distributed unit (DU). Unlike idealized settings with perfect instantaneous CSI, real-world deployments are subject to quantization constraints and feedback delay, both of which significantly impact CJT performance.

A. Codebook-Based CSI Feedback

We assume standardized codebook-based CSI feedback, where each UE reports quantized channel information using a Type-II codebook [24]. In distributed MIMO systems, conventional CSI feedback methods designed for co-located antenna arrays are not directly applicable due to the spatial separation of radio units (RUs). To address this, we adopt the CB2 distributed CSI feedback structure from [25] in which per-RU spatial bases are selected independently and combined to form a global representation. This enables CJT precoder construction using quantized CSI while preserving the distributed spatial structure.

B. CSI Aging and Feedback Delay

Let P_{fb} denote the feedback delay between CSI estimation at the UE and its use at the DU. Due to mobility, the channel evolves over this delay, resulting in a mismatch between the precoder and the instantaneous channel.

At time t , the DU constructs the downlink precoder using CSI corresponding to time $t - P_{fb}$. This temporal mismatch degrades coherent combining gains, particularly in MD-MIMO systems where multiple distributed links must align in phase. To mitigate CSI aging, the UE predicts the future channel state corresponding to time $t + P_{fb}$. This prediction is then used to calculate codebook-based CSI feedback.

IV. PRELIMINARIES

In this section, we review two fundamental building blocks used in the channel prediction method introduced in this work: linear state-space estimation via Kalman filtering and Windowed Echo State Network (WESN). This section is kept general and independent of any specific application.

A. Kalman Filtering

A discrete-time linear state-space model can be expressed as:

$$\mathbf{x}_t = \mathbf{F}\mathbf{x}_{t-1} + \mathbf{w}_t, \quad (4)$$

$$\mathbf{y}_t = \mathbf{H}\mathbf{x}_t + \mathbf{n}_t, \quad (5)$$

where:

- $\mathbf{x}_t \in \mathbb{C}^{N_x \times 1}$ is the latent state vector,
- $\mathbf{y}_t \in \mathbb{C}^{N_y \times 1}$ is the observation vector,
- $\mathbf{F} \in \mathbb{C}^{N_x \times N_x}$ is the state transition matrix,
- $\mathbf{H} \in \mathbb{C}^{N_y \times N_x}$ is the observation matrix,
- $\mathbf{w}_t \sim \mathcal{CN}(\mathbf{0}, \mathbf{Q})$, $\mathbf{Q} \in \mathbb{C}^{N_x \times N_x}$,
- $\mathbf{n}_t \sim \mathcal{CN}(\mathbf{0}, \mathbf{R})$, $\mathbf{R} \in \mathbb{C}^{N_y \times N_y}$.

The Kalman filter provides the minimum mean-square error (MMSE) estimate of the state through the recursion:

$$\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1|t-1}, \quad (6)$$

$$\mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^H + \mathbf{Q}, \quad (7)$$

$$\mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{H}^H (\mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}^H + \mathbf{R})^{-1}, \quad (8)$$

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t (\mathbf{y}_t - \mathbf{H}\hat{\mathbf{x}}_{t|t-1}), \quad (9)$$

$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\mathbf{H})\mathbf{P}_{t|t-1}, \quad (10)$$

where:

- $\hat{\mathbf{x}}_{t|t-1}$ and $\hat{\mathbf{x}}_{t|t}$ are the predicted and updated state estimates,
- $\mathbf{P}_{t|t-1}, \mathbf{P}_{t|t} \in \mathbb{C}^{N_x \times N_x}$ are the corresponding error covariance matrices,
- $\mathbf{K}_t \in \mathbb{C}^{N_x \times N_y}$ is the Kalman gain.

This recursive structure defines a linear dynamical system that maps past observations to future state estimates.

B. Windowed Echo State Network

Echo State Network (ESN) belongs to the class of reservoir computing models for temporal sequence processing. In this work, we use a Windowed ESN (WESN) that explicitly incorporates short-term temporal context. An ESN architecture is illustrated in Fig. 2.

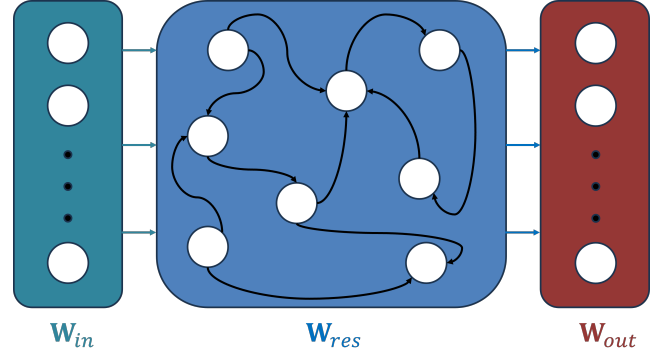


Fig. 2. Vanilla ESN architecture

1) *Windowed Input Representation*: Let $\mathbf{u}(t) \in \mathbb{C}^{N_u \times 1}$ denote the input sequence. A causal sliding window of length L_w is constructed as:

$$\tilde{\mathbf{u}}(t) = [\mathbf{u}(t)^T \quad \mathbf{u}(t-1)^T \quad \cdots \quad \mathbf{u}(t-L_w+1)^T]^T, \quad (11)$$

where $\tilde{\mathbf{u}}(t) \in \mathbb{C}^{(L_w N_u) \times 1}$.

This representation allows the network to access recent temporal history explicitly.

2) *Reservoir Dynamics with Skip Connections*: Let the reservoir state be $\mathbf{s}(t) \in \mathbb{C}^{N_n \times 1}$, where N_n is the number of reservoir neurons. The state evolves according to:

$$\mathbf{s}(t) = \phi(\mathbf{W}_{res}\mathbf{s}(t-1) + \mathbf{W}_{in}\tilde{\mathbf{u}}(t)), \quad (12)$$

where:

- $\mathbf{W}_{res} \in \mathbb{C}^{N_n \times N_n}$ is the recurrent weight matrix,
- $\mathbf{W}_{in} \in \mathbb{C}^{N_n \times (L_w N_u)}$ is the input projection matrix,
- $\phi(\cdot)$ is an element-wise nonlinear activation function.

To improve information flow and stabilize learning, we augment the model with a linear skip connection from the input to the output.

3) *Readout and Training*: The output $\hat{\mathbf{y}}(t) \in \mathbb{C}^{N_y \times 1}$ is computed as:

$$\hat{\mathbf{y}}(t) = \mathbf{W}_{out} \begin{bmatrix} \mathbf{s}(t) \\ \tilde{\mathbf{u}}(t) \end{bmatrix}, \quad (13)$$

where:

- $\mathbf{W}_{out} \in \mathbb{C}^{N_y \times (N_n + L_w N_u)}$ is the readout matrix.

Only the readout weights $\mathbf{W}_{out}$ are trained, typically using ridge regression:

$$\mathbf{W}_{out} = \mathbf{T}\mathbf{Z}^H (\mathbf{Z}\mathbf{Z}^H + \lambda\mathbf{I})^{-1}, \quad (14)$$

where:

- $\mathbf{Z}$ stacks the augmented features $[\mathbf{s}(t)^T, \tilde{\mathbf{u}}(t)^T]^T$ over time,
- $\mathbf{T}$ stacks the corresponding target outputs,
- $\lambda > 0$ is a regularization parameter.

The reservoir and input weights are fixed after initialization, while the readout is trained using observed data. This structure enables efficient, low-complexity online adaptation.

V. CONFIGURED WESN FOR CHANNEL PREDICTION

We introduce a strategy for configuring the recurrent and input weights of the WESN for channel prediction using statistics of Kalman filtering. The motivation follows the broader configured-reservoir viewpoint: instead of randomly generating the recurrent dynamics, the WESN weights can be configured using the statistics of a target transfer-function family. This idea was developed for MIMO receive processing in [22], and we specialize it to channel prediction. The WESN configuration and training pipeline is shown in Fig. 3.

A natural target dynamical system is the Kalman predictor. For linear Gaussian state-space models, the Kalman filter provides the MMSE predictor when the state-transition matrix and the process/observation noise covariances are known exactly. However, this is also a limitation in practice, since prediction performance depends on the accuracy of the assumed model parameters. In the introduced approach, we therefore use the statistics of a Kalman-based predictor only to configure the WESN reservoir, while still learning the output weights from empirical data. In this way, the recurrent dynamics inherit model-based structure, whereas the trained readout compensates for model mismatch and imperfect knowledge of the underlying dynamics.

For analytical tractability, we begin with a linearized view of the WESN and focus first on the recurrent component with parallel non-interconnected neurons, following the configured-reservoir perspective in [22]. As argued in [22], this linearized setting exposes the underlying signal-processing structure of the reservoir while preserving the essential role of weight configuration. The nonlinear activation, skip connections, and final trained readout can then be viewed as additional mechanisms that improve robustness and approximation accuracy beyond this core linear dynamical model.

Consider a bank of N_n parallel recurrent neurons driven by an input $\tilde{\mathbf{u}}(t) \in \mathbb{C}^{N_{\text{in}} \times 1}$. We consider a linearized WESN with non-interconnected neurons, so that the recurrent weight matrix is diagonal:

$$\mathbf{W}_{\text{res}} = \text{diag}(w_{\text{res},1}, \dots, w_{\text{res},N_n}). \quad (15)$$

Let $\mathbf{w}_{\text{in},i}^T$ denote the i th row of the input weight matrix $\mathbf{W}_{\text{in}} \in \mathbb{C}^{N_n \times N_{\text{in}}}$. Then, under linear activation, the state recursion of the i th neuron is

$$s_i[t] = w_{\text{res},i} s_i[t-1] + \mathbf{w}_{\text{in},i}^T \tilde{\mathbf{u}}(t). \quad (16)$$

Taking the z -transform, the input-output transfer function of the i th neuron is

$$H_i(z) = \frac{\mathbf{w}_{\text{in},i}^T}{1 - w_{\text{res},i} z^{-1}}. \quad (17)$$

Thus, each neuron implements a first-order IIR mode whose pole is given by the corresponding diagonal entry of $\mathbf{W}_{\text{res}}$, and whose input coupling is determined by the corresponding row of $\mathbf{W}_{\text{in}}$. Since the neurons are not interconnected, the reservoir spans a linear subspace generated by these first-order modes, and the final readout forms a linear combination of them.

Let $\mathbf{W}_{\text{out}} \in \mathbb{C}^{N_{\text{out}} \times N_n}$ denote the readout matrix. Then the overall transfer function of the linearized WESN can be written as

$$\mathbf{T}_{\text{WESN}}(z) = \sum_{i=1}^{N_n} \frac{\mathbf{W}_{\text{out}}[:,i] \mathbf{w}_{\text{in},i}^T}{1 - w_{\text{res},i} z^{-1}}, \quad (18)$$

where $\mathbf{W}_{\text{out}}[:,i]$ denotes the i th column of $\mathbf{W}_{\text{out}}$.

Equation (18) shows that the WESN realizes a sum of first-order IIR components, each corresponding to a neuron. Therefore, configuring the reservoir requires selecting the poles $\{w_{\text{res},i}\}$ and input couplings $\{\mathbf{w}_{\text{in},i}\}$ such that the WESN can approximate a desired target transfer-function family.

A natural choice is the Kalman predictor. For linear Gaussian state-space models, the Kalman filter provides the MMSE predictor when the state-transition matrix and the process/observation noise covariances are known exactly. However, this model-dependent MMSE property is also a limitation in practice, since prediction performance depends on the accuracy of the assumed model parameters. Accordingly, our strategy is to use the transfer-function statistics of a Kalman-based predictor to determine the poles and input couplings of the WESN reservoir, while still learning the final output combination from empirical data.

For a given UE-RU pair (k, g) , let

$$\mathbf{h}_{k,g}[t] \triangleq \text{vec}(\mathbf{H}_{k,g}[t]) \in \mathbb{C}^{M_k N_g \times 1}, \quad (19)$$

$$\tilde{\mathbf{h}}_{k,g}[t] \triangleq \text{vec}(\tilde{\mathbf{H}}_{k,g}[t]) \in \mathbb{C}^{M_k N_g \times 1}, \quad (20)$$

denote the true and noisy channel vectors, respectively. To capture temporal correlation, we use an autoregressive model of order P :

$$\mathbf{h}_{k,g}[t+1] = \sum_{\ell=1}^P \mathbf{A}_{\ell} \mathbf{h}_{k,g}[t+1-\ell] + \mathbf{w}_{k,g}[t], \quad (21)$$

where $\mathbf{A}_{\ell} \in \mathbb{C}^{M_k N_g \times M_k N_g}$ and

$$\mathbf{w}_{k,g}[t] \sim \mathcal{CN}(\mathbf{0}, \mathbf{Q}). \quad (22)$$

The corresponding observation model is

$$\tilde{\mathbf{h}}_{k,g}[t] = \mathbf{h}_{k,g}[t] + \mathbf{n}_{k,g}[t], \quad \mathbf{n}_{k,g}[t] \sim \mathcal{CN}(\mathbf{0}, \mathbf{R}). \quad (23)$$

The AR order P determines the memory depth of the model-based predictor. It also guides the learning architecture, since a predictor of order P depends on the most recent P channel observations or states. This motivates linking the WESN window length L_w to the effective temporal memory of the AR model with $L_w = P$.

To write (21) in first-order form, define the augmented state

$$\mathbf{z}_{k,g}[t] \triangleq [\mathbf{h}_{k,g}[t]^T \quad \mathbf{h}_{k,g}[t-1]^T \quad \dots \quad \mathbf{h}_{k,g}[t-P+1]^T]^T, \quad (24)$$

where $\mathbf{z}_{k,g}[t] \in \mathbb{C}^{PM_k N_g \times 1}$.

Then

$$\mathbf{z}_{k,g}[t+1] = \mathbf{F}_{\text{aug}} \mathbf{z}_{k,g}[t] + \mathbf{w}_{k,g}^{\text{aug}}[t], \quad (25)$$

$$\tilde{\mathbf{h}}_{k,g}[t] = \mathbf{C}_{\text{obs}} \mathbf{z}_{k,g}[t] + \mathbf{n}_{k,g}[t], \quad (26)$$

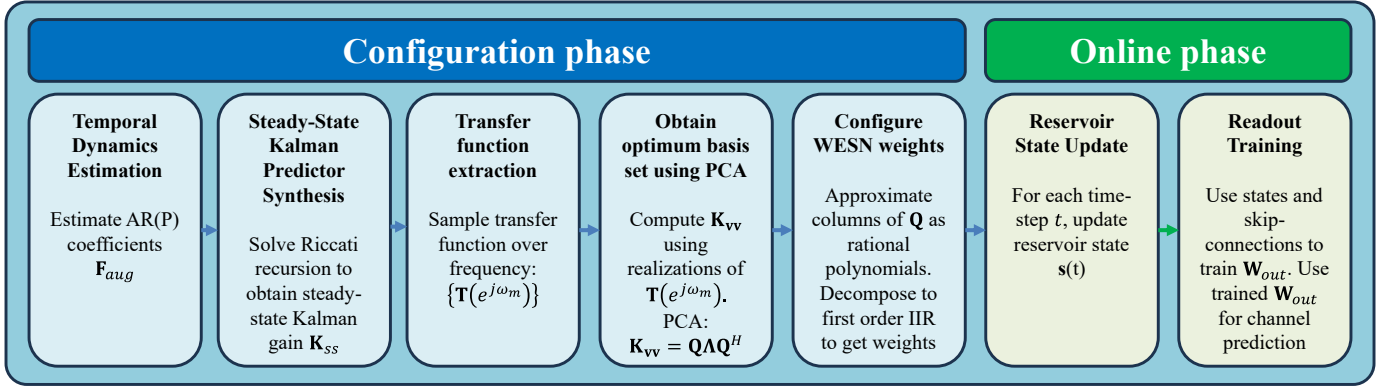


Fig. 3. Configured WESN channel prediction pipeline.

where $\mathbf{F}_{\text{aug}} \in \mathbb{C}^{PM_k N_g \times PM_k N_g}$ is the standard companion-form augmented state-transition matrix induced by $\{\mathbf{A}_\ell\}_{\ell=1}^P$, and

$$\mathbf{C}_{\text{obs}} = [\mathbf{I}_{M_k N_g} \quad \mathbf{0} \quad \cdots \quad \mathbf{0}] \in \mathbb{C}^{M_k N_g \times PM_k N_g} \quad (27)$$

selects the first block of the augmented state.

For this model, the full Kalman predictor is time-varying because the Kalman gain $\mathbf{K}_t$ evolves with time. Hence, the induced predictor does not in general admit a single fixed transfer function. Since the configured-reservoir method requires a fixed target operator, we instead use the steady-state Kalman predictor as the target system.

Under the usual assumptions of a time-invariant model together with stabilizability and detectability, the Kalman gain converges to a steady-state value [26]:

$$\mathbf{K}_t \rightarrow \mathbf{K}_{\text{ss}}. \quad (28)$$

A convenient way to define $\mathbf{K}_{\text{ss}}$ is through the steady-state Riccati fixed point. Let $\mathbf{P}_{\text{ss}}^-$ denote the steady-state a priori error covariance. Then

$$\mathbf{S}_{\text{ss}} = \mathbf{C}_{\text{obs}} \mathbf{P}_{\text{ss}}^- \mathbf{C}_{\text{obs}}^H + \mathbf{R}, \quad (29)$$

$$\mathbf{K}_{\text{ss}} = \mathbf{P}_{\text{ss}}^- \mathbf{C}_{\text{obs}}^H \mathbf{S}_{\text{ss}}^{-1}, \quad (30)$$

with

$$\mathbf{P}_{\text{ss}}^+ = \mathbf{P}_{\text{ss}}^- - \mathbf{K}_{\text{ss}} \mathbf{C}_{\text{obs}} \mathbf{P}_{\text{ss}}^-, \quad (31)$$

$$\mathbf{P}_{\text{ss}}^- = \mathbf{F}_{\text{aug}} \mathbf{P}_{\text{ss}}^+ \mathbf{F}_{\text{aug}}^H + \mathbf{Q}_{\text{aug}}. \quad (32)$$

We obtain the steady-state solution by iterating the Riccati recursion until convergence and then evaluating (30).

Using the standard prediction-update form, the resulting steady-state predictor is

$$\hat{\mathbf{z}}_{k,g}[t+1|t] = \mathbf{A}_p \hat{\mathbf{z}}_{k,g}[t|t-1] + \mathbf{B}_p \tilde{\mathbf{h}}_{k,g}[t], \quad (33)$$

where

$$\mathbf{A}_p \triangleq \mathbf{F}_{\text{aug}} (\mathbf{I} - \mathbf{K}_{\text{ss}} \mathbf{C}_{\text{obs}}), \quad (34)$$

$$\mathbf{B}_p \triangleq \mathbf{F}_{\text{aug}} \mathbf{K}_{\text{ss}}. \quad (35)$$

The one-step-ahead channel prediction is then obtained by selecting the first block of the predicted state:

$$\hat{\mathbf{h}}_{k,g}[t+1|t] = \mathbf{C}_{\text{obs}} \hat{\mathbf{z}}_{k,g}[t+1|t]. \quad (36)$$

Therefore, the steady-state predictor defines the Linear Time Invariant (LTI) transfer function

$$\mathbf{T}_{k,g}(z) = \mathbf{C}_{\text{obs}} (\mathbf{I} - \mathbf{A}_p z^{-1})^{-1} \mathbf{B}_p, \quad (37)$$

which maps the noisy channel observations $\tilde{\mathbf{h}}_{k,g}[t]$ to the one-step-ahead channel prediction $\hat{\mathbf{h}}_{k,g}[t+1|t]$.

Rather than using only a single transfer function, we build a family of steady-state predictor transfer functions from multiple channel-history windows. For each window index s , we estimate the AR model, construct $\mathbf{F}_{\text{aug}}^{(s)}$, solve for the corresponding steady-state gain $\mathbf{K}_{\text{ss}}^{(s)}$, and evaluate

$$\mathbf{T}_{k,g}^{(s)}(z) = \mathbf{C}_{\text{obs}} (\mathbf{I} - \mathbf{A}_p^{(s)} z^{-1})^{-1} \mathbf{B}_p^{(s)}. \quad (38)$$

Each realization is sampled over a frequency grid $\{\omega_m\}_{m=1}^{N_\omega}$:

$$\mathbf{T}_{k,g}^{(s)}(e^{j\omega_m}), \quad m = 1, \dots, N_\omega. \quad (39)$$

Each sampled transfer matrix is vectorized as

$$\mathbf{v}^{(s)}(\omega_m) = \text{vec}(\mathbf{T}_{k,g}^{(s)}(e^{j\omega_m})), \quad (40)$$

and stacked across frequencies and predictor realizations. This produces a sample matrix of transfer-function vectors.

From these samples, we form the empirical covariance

$$\mathbf{K}_{vv} = \mathbb{E}[\mathbf{v}\mathbf{v}^H]. \quad (41)$$

We then perform eigendecomposition

$$\mathbf{K}_{vv} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^H, \quad (42)$$

where the columns of $\mathbf{Q}$ are the eigenvectors and the diagonal entries of $\mathbf{\Lambda}$ are the corresponding eigenvalues. Retaining the M dominant eigenvectors gives

$$\mathbf{Q}_M = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \cdots \quad \mathbf{q}_M], \quad (43)$$

which spans the principal subspace of the steady-state Kalman predictor family. Thus, instead of forcing the WESN to reproduce the entire transfer-function family exactly, we configure it to represent its dominant covariance modes. In this work, the number of retained modes M is selected heuristically as

a hyperparameter. Each dominant eigenvector $\mathbf{q}_m$ is next interpreted as a sampled vectorized frequency response. Specifically, after reshaping $\mathbf{q}_m$ across the frequency grid, we obtain a vector-valued function

$$\mathbf{q}_m(e^{j\omega}) \in \mathbb{C}^{(M_k N_g)^2 \times 1}, \quad (44)$$

whose entries correspond to the vectorized transfer matrix at frequency ω .

We then approximate each mode using a vector rational polynomial with shared poles:

$$\mathbf{q}_m(e^{j\omega}) \approx \sum_{\ell=1}^K \frac{\mathbf{c}_{m,\ell}}{1 - p_{m,\ell} e^{-j\omega}}, \quad (45)$$

where $p_{m,\ell} \in \mathbb{C}$ are the poles and $\mathbf{c}_{m,\ell} \in \mathbb{C}^{(M_k N_g)^2 \times 1}$ are the corresponding residue vectors.

The approximation in (45) directly matches the transfer-function structure of the linearized WESN in (18). Specifically, each term in (45) has the same first-order IIR form as the basis functions generated by individual reservoir neurons.

For each mode m and term ℓ , we associate a non-interconnected neuron whose recurrent weight is given by the corresponding pole:

$$w_{\text{res},m,\ell} = p_{m,\ell}. \quad (46)$$

Similarly, the corresponding row of the input weight matrix is constructed from the residue vector $\mathbf{c}_{m,\ell}$:

$$\mathbf{w}_{\text{in},m,\ell}^T = \mathbf{c}_{m,\ell}. \quad (47)$$

Under this construction, each (m, ℓ) pair defines a first-order IIR neuron with transfer function

$$H_{m,\ell}(z) = \frac{\mathbf{w}_{\text{in},m,\ell}^T}{1 - w_{\text{res},m,\ell} z^{-1}}, \quad (48)$$

which corresponds to one term in the pole-residue expansion of the dominant covariance modes.

Stacking all such components yields a non-interconnected reservoir with

$$\mathbf{W}_{\text{res}} = \text{diag}\left(\{p_{m,\ell}\}_{m=1,\ell=1}^{M,K}\right), \quad (49)$$

and

$$\mathbf{W}_{\text{in}} = \begin{bmatrix} \mathbf{w}_{\text{in},1,1}^T \\ \mathbf{w}_{\text{in},1,2}^T \\ \vdots \\ \mathbf{w}_{\text{in},M,K}^T \end{bmatrix}. \quad (50)$$

Thus, the configured WESN reservoir consists of MK parallel first-order IIR neurons, whose poles $\{p_{m,\ell}\}$ and input couplings $\{\mathbf{w}_{\text{in},m,\ell}\}$ are directly determined from the rational approximation of the dominant Kalman-predictor modes. In practice, the poles are further scaled to satisfy a prescribed spectral-radius constraint, and the input weights are scaled by an input-gain factor before running the reservoir recursion.

The skip connection in the WESN output layer provides an additional finite impulse response (FIR) component beyond the IIR reservoir dynamics. Hence, the configured WESN combines two complementary parts: an IIR part arising from the pole-residue realization of the dominant Kalman-predictor

modes, and an FIR part arising from the direct windowed input skip path. The latter adds flexibility for compensating residual mismatch and for capturing predictor components that are not well represented by the truncated all-pole recurrent part alone.

Finally, once the recurrent and input weights have been configured from the steady-state predictor statistics, only the readout weights are learned from data through ridge regression. This hybrid design combines the strengths of model-based and data-driven prediction: the configured reservoir embeds the structure of a Kalman-filtering based predictor, while the learned readout reduces reliance on exact knowledge of $\mathbf{F}_{\text{aug}}$, $\mathbf{Q}$, and $\mathbf{R}$ by adapting directly to the empirical data observed during operation.

Although the above discussion is written for a single MIMO channel matrix, the implementation operates on the full OFDM channel tensor. Specifically, for each UE-RU link, the channel at time t consists of matrices $\mathbf{H}_{k,g}[t, f, o]$ for subcarrier f and OFDM symbol o . The configured reservoir weights are shared across these time-frequency resource elements, and the readout is trained using samples from all subcarriers and OFDM symbols jointly.

To see this, define

$$\mathcal{B} \triangleq \{(f, o) : f = 1, \dots, N_f, o = 1, \dots, N_{\text{sym}}\}, \quad (51)$$

and let

$$\mathbf{u}_{k,g}^{(f,o)}[t] \quad (52)$$

denote the windowed input constructed from the past observations of $\mathbf{H}_{k,g}[t, f, o]$. During online prediction, the reservoir is rolled out independently for each $(f, o) \in \mathcal{B}$ using the same configured $\mathbf{W}_{\text{res}}$ and $\mathbf{W}_{\text{in}}$. The readout is then obtained from a single ridge-regression problem whose feature and target matrices stack all training pairs over resource elements:

$$\mathbf{W}_{\text{out}} = \mathbf{T}_{\text{OFDM}} \mathbf{Z}_{\text{OFDM}}^H (\mathbf{Z}_{\text{OFDM}} \mathbf{Z}_{\text{OFDM}}^H + \lambda \mathbf{I})^{-1}. \quad (53)$$

Here, $\mathbf{Z}_{\text{OFDM}}$ stacks the augmented WESN features generated from $\{\mathbf{u}_{k,g}^{(f,o)}[t]\}$ over all available t and all $(f, o) \in \mathcal{B}$, while $\mathbf{T}_{\text{OFDM}}$ stacks the corresponding one-step-ahead channel targets. Thus, although the model derivation is presented on a per-subcarrier basis for notational clarity, the online readout exploits the full OFDM grid by solving one least-squares problem over all subcarriers and OFDM symbols.

Each UE applies the introduced predictor to estimate the future channel $\mathbf{H}_k(t + P_{\text{fb}})$ based on past observations. The predicted channel is then used to generate CSI feedback using the codebook-based framework described in Section III.

The DU constructs the CJT precoder using the received predicted CSI. In this work, we employ a zero-forcing (ZF) precoder based on the quantized predicted channel [24]. By compensating for CSI aging, the introduced prediction framework enables more accurate precoder design and improved coherent combining across distributed RUs.

VI. COMPLEXITY ANALYSIS

We compare the computational complexity of the configured WESN predictor and the Kalman-filtering baseline for one UE-RU link. Let

$$D \triangleq M_k N_g, \quad (54)$$

and define the number of OFDM resource elements per slot as

$$B \triangleq N_f N_{\text{sym}}. \quad (55)$$

Let T denote the channel-history buffer length and P the AR order. For the configured WESN, we set $L_w = P$.

A. Kalman Predictor

The Kalman predictor consists of two stages: AR model estimation and per-resource-element Kalman recursion.

The AR model is estimated jointly across all subcarriers and OFDM symbols using least squares with feature dimension PD and $(T - P)B$ samples. This requires

$$\mathcal{O}((T - P)B(PD)^2 + (PD)^3). \quad (56)$$

After model estimation, the Kalman recursion is executed independently for each resource element. With augmented state dimension PD , each time update requires $\mathcal{O}((PD)^3)$ operations. Over $(T - P)$ time steps and B resource elements, this yields

$$\mathcal{O}(B(T - P)(PD)^3). \quad (57)$$

Thus, the overall Kalman complexity is

$$\mathcal{O}((T - P)B(PD)^2 + (PD)^3 + B(T - P)(PD)^3). \quad (58)$$

B. Configured WESN (Online Phase)

The configured WESN uses $N_n = MK$ non-interconnected neurons and input dimension $N_{\text{in}} = PD$. The readout feature dimension is

$$N_z = MK + PD. \quad (59)$$

Each neuron performs a dot product of length PD . Over MK neurons, T time steps, and B resource elements:

$$\mathcal{O}(TBMKPD). \quad (60)$$

For readout training, a ridge-regression problem is solved over $(T - 1)B$ samples with feature dimension N_z , requiring

$$\mathcal{O}((T - 1)BN_z^2 + N_z^3). \quad (61)$$

Applying the readout across all resource elements requires

$$\mathcal{O}(BDN_z). \quad (62)$$

The total online WESN complexity is therefore

$$\mathcal{O}(TBMKPD + (T - 1)B(MK + PD)^2 + (MK + PD)^3 + BD(MK + PD)). \quad (63)$$

Table I provides a numerical evaluation of the per-link computational complexity in terms of the total number of complex multiplications required by each predictor under the listed parameter settings. The results highlight how the relative complexity of the two predictors scales with the MIMO dimension, with the configured WESN scaling more efficiently with higher MIMO orders.

TABLE I
PER-LINK COMPLEXITY COMPARISON

Antenna Configuration	Kalman Filtering	Configured WESN (online)
2×2	1.24×10^7	1.14×10^7
4×2	9.36×10^7	2.35×10^7
4×4	7.27×10^8	6.14×10^7
8×8	4.55×10^{10}	6.32×10^8
Parameters: $N_f = 512$, $N_{\text{sym}} = 14$, $T = 5$, $P = 2$, $M = 3$, $K = 3$		

VII. RESULTS AND DISCUSSION

A. Experiment Design

To evaluate the performance of the MD-MIMO CJT architecture with the introduced channel prediction strategy, we conducted comprehensive simulation evaluation. The simulation framework we adopt combines the ns-3 network simulator with the Sionna link-level simulator to emulate the radio propagation environment and the PHY layer operations.

To be specific, we use the ns-3 simulator to model the network environment, incorporating the 3GPP Vehicle-to-Vehicle (V2V) Urban channel model as specified in 3GPP TR 37.885 [27]. We randomly placed a number of distributed RUs within a 100 meter circle centered around the DU. The UEs were placed in another circle of radius 100 meters located 1000 meters away from the DU. We assume RU 0 is equipped with 4 antennas and the rest of the RUs have 2 antennas each. We also assume that each UE has 2 antennas. In our evaluation, all RUs, the DU, and all UEs are assumed to be mobile. The mobility of each device was divided into a common component and an individual component. Specifically, the RUs were modeled to have a common speed in one direction, with an additional smaller individual speed in a random direction. The same modeling approach was applied to the UEs. To evaluate channel prediction strategies across a range of radio propagation conditions, we place the RUs and UEs at random locations within the 100 meter radius regions described above. Each such random spatial realization of the network topology is referred to as a “drop”. For each drop, we generate channel realizations over 100 slots, where each slot duration is 1 ms. We repeat this process for 100 independent drops, thereby evaluating the system over a diverse set of channel conditions. Each drop exhibits distinct large-scale and small-scale fading characteristics, including variations in path loss, shadowing, and line-of-sight (LOS) versus non-line-of-sight (NLOS) conditions. This approach enables robust performance evaluation by averaging over both spatial randomness and temporal channel dynamics.

The Sionna link-level simulator [28] is employed to obtain the link level performance metrics. Sionna was configured to transmit pilot symbols over the channels modeled by ns-3, enabling the emulation of realistic channel estimation processes at the receiver. The Kronecker-style scattered pilot pattern was used in the evaluation. Each UE was configured to perform Linear Minimum Mean Square Error (LMMSE) channel estimation. The simulation parameters for the evaluations are summarized in Table II.

TABLE II
SIMULATOR PARAMETERS

System Parameters	Values
Center Frequency	3.5 GHz
Subcarrier Spacing	15 kHz
Number of Subcarriers	512
Channel Feedback Delay	4 ms
Buffer Size (B_{fb})	5 slots
AR Order (P)	2
RU Tx Power	35 dBm (RU 0), 25 dBm (other)
Number of UEs	4

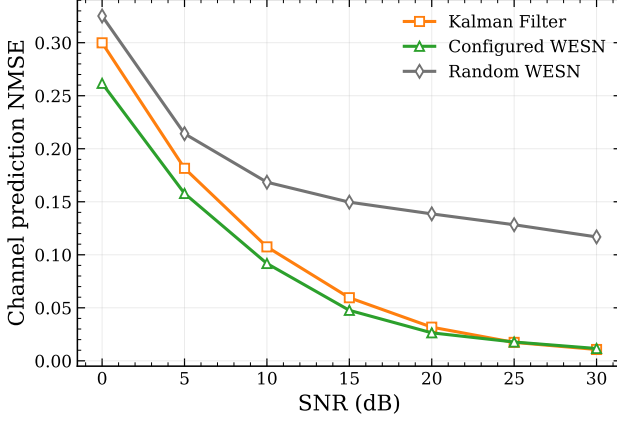


Fig. 4. Channel prediction NMSE vs SNR at 10 km/h

B. Results

Fig. 4 evaluates the introduced WESN weight configuration strategy in a simplified point-to-point channel prediction setting. In this experiment, we isolate the impact of temporal prediction by directly operating on small-scale fading realizations extracted from the ns-3 simulations described in Section VII-A. The mobility for this evaluation is fixed at 10 km/h. Specifically, we take the ground-truth channel realizations $\mathbf{H}_{GT}[t]$ from the ns-3 traces and construct noisy observations according to

$$\mathbf{H}_{obs}[t] = \mathbf{H}_{GT}[t] + \mathbf{n}[t], \quad (64)$$

where $\mathbf{n}[t] \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I})$ is additive Gaussian noise corresponding to a target SNR. In this controlled setup, the noise variance σ^2 is assumed to be perfectly known, allowing us to isolate the effect of imperfect knowledge of channel dynamics.

Following the framework developed in Section V, we fit a linear state-space model to the observed channel sequence. The state transition matrix $\mathbf{F}$ is not assumed known and is instead estimated directly from data. A buffer of past noisy channel observations is used to form a least-squares problem, from which the autoregressive coefficients (and hence $\mathbf{F}$) are obtained. This estimated $\mathbf{F}$ is then used to construct both the Kalman predictors and the transfer-function statistics used for configuring the WESN reservoir.

Fig. 4 shows the resulting channel prediction NMSE as a function of SNR. From the figure, it can be observed that when the channel dynamics are imperfectly known through

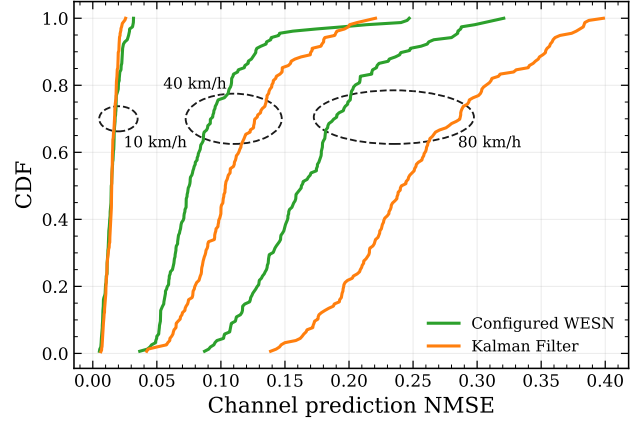


Fig. 5. Empirical CDF of channel prediction NMSE at mobility speeds of 10, 40, and 80 km/h

the estimated $\mathbf{F}$, the performance of Kalman filtering degrades, particularly in the low SNR regime where the estimation of $\mathbf{F}$ becomes less reliable. In contrast, the configured WESN is able to compensate for this model mismatch through its learned readout, resulting in significant performance gains over Kalman filtering at low to medium SNR. At high SNR, $\mathbf{F}$ can be estimated accurately, which leads to similar NMSEs for both configured WESN and Kalman filtering. Additionally, the configured WESN consistently outperforms a WESN with randomly initialized reservoir weights. This highlights the benefit of embedding model-based structure into the reservoir.

Fig. 5 evaluates the introduced channel prediction strategy under a realistic link-level simulation of the MD-MIMO system described in Section VII-A. This experiment incorporates both large-scale and small-scale channel effects, as well as practical channel estimation errors. Specifically, pilot symbols are transmitted over the ns-3 generated channels, and the received signals are used to perform LMMSE channel estimation at the UE. This introduces both thermal noise and estimation errors into the observed channel sequence. As a result, both the state transition matrix $\mathbf{F}$ and the observation noise covariance are estimated from these noisy channel estimates rather than being known a priori.

Using this setup, we evaluate one-step-ahead channel prediction performance across multiple mobility scenarios. Fig. 5 shows the empirical CDF of the resulting NMSE values aggregated across UEs, RUs, and time instances.

From the figure, we observe that the configured WESN provides a consistent improvement over standard Kalman filtering across all mobility regimes. The gains are particularly pronounced at higher mobilities, where the temporal dynamics become more difficult to estimate accurately and model mismatch becomes more severe.

These results further validate the benefit of incorporating model-based structure into the reservoir, particularly in realistic settings where both the channel dynamics and observation statistics must be estimated. In addition to the observed prediction accuracy gains, the configured WESN also offers a significant advantage in terms of online computational complexity. As detailed in Section VI, the configured WESN operates

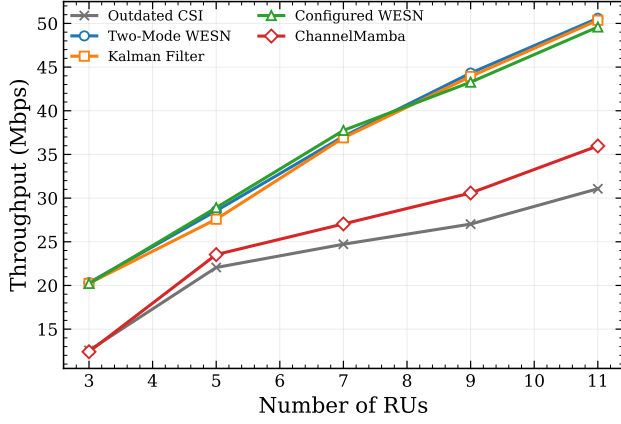


Fig. 6. Throughput vs number of RUs at 10 km/h under the time-based split.

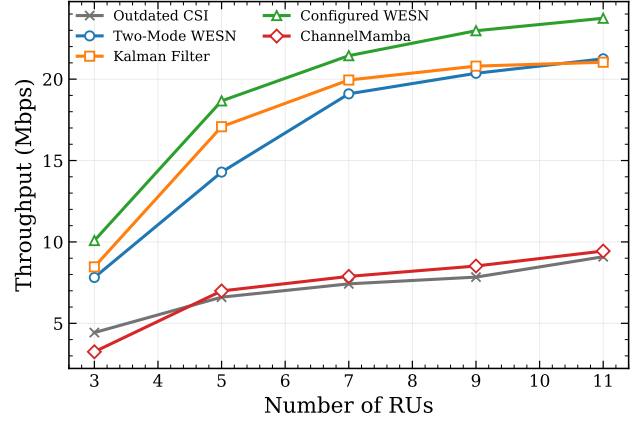


Fig. 8. Throughput vs number of RUs at 80 km/h under the time-based split.

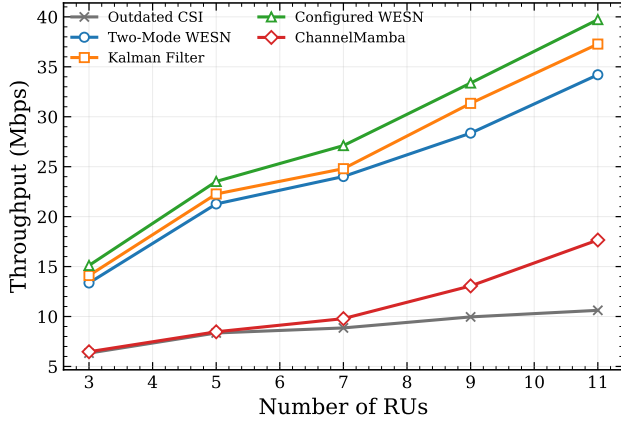


Fig. 7. Throughput vs number of RUs at 40 km/h under the time-based split.

with substantially lower computational cost, particularly as the MIMO dimension increases. Consequently, the configured WESN approach not only improves prediction accuracy under model mismatch, but also provides a more scalable and practical solution for real-time MD-MIMO deployments.

Fig. 6, Fig. 7 and Fig. 8 evaluate the end-to-end impact of channel prediction on system performance by plotting the achieved downlink throughput as a function of the number of distributed RUs at 10, 40, and 80 km/h, respectively. These results quantify how improvements in channel prediction NMSE translate into meaningful system-level gains.

The predicted channel is used to generate Type-II codebook-based CSI feedback at the UE. The DU then reconstructs the channel from the quantized feedback and computes a zero-forcing (ZF) precoder for CJT across the distributed RUs. The quality of channel prediction directly impacts the accuracy of the precoder, and consequently the effectiveness of inter-user interference suppression. For these results, we utilize a link-adaptation algorithm to maximize throughput over a selection of Modulation and Coding Schemes (MCS). These figures show the coded throughput after LDPC decoding.

We include an online learning-based baseline, namely the two-mode WESN channel predictor introduced in [5]. This method relies on randomly initialized reservoir dynamics,

but preserves the structure of the MIMO channel matrix during prediction through bilinear state updates. In contrast, the configured WESN leverages structure derived from the Kalman predictor, resulting in a more efficient representation of the underlying channel dynamics. As a result, the configured WESN is observed to outperform both Kalman filtering and the two-mode WESN baseline.

In addition to the Kalman filtering and reservoir computing baselines, we also include ChannelMamba [15] as a learning-based benchmark in Fig. 6-Fig. 8. ChannelMamba is a recent state-of-the-art deep learning approach for channel prediction that leverages a Mamba-based selective state-space model to capture long-range temporal dependencies in high-mobility MIMO channels. It is trained in an end-to-end fashion using large offline datasets and has demonstrated strong performance in prior work. Since ChannelMamba is an offline training-based method, its evaluation requires an explicit separation between training and testing data. For Fig. 6-Fig. 8, we train the model using the first 50% of the time slots from all drops and evaluate it on the remaining 50% of time slots within the same drops. The 50/50 split is chosen for a fair comparison with the configured WESN, where the configuration phase and the subsequent online training/testing phase also operate over a 50-50 split. From Fig. 6-Fig. 8, we observe that all online approaches consistently outperform ChannelMamba. This highlights the advantage of methods that adapt directly to the observed channel during operation, rather than relying solely on offline training.

From Fig. 6, we observe that all prediction methods benefit from an increasing number of RUs due to enhanced spatial degrees of freedom. At lower mobility (10 km/h), the performance gap between the online predictors is relatively modest, indicating that temporal channel dynamics can be captured reasonably well even with imperfect models.

In contrast, Fig. 7 and Fig. 8 show that at higher mobility, the choice of predictor has a much more pronounced impact. The configured WESN consistently achieves higher throughput compared to the baselines. This improvement can be attributed to more accurate channel prediction under rapidly varying channel conditions, which enables more precise precoder

construction and improved interference cancellation.

Next, Fig. 9, Fig. 10, and Fig. 11 consider a more challenging spatial generalization setting. Instead of splitting the dataset along the time dimension within each drop, we partition the dataset across drops. Specifically, ChannelMamba is trained using the first 50% of drops, while the configured WESN uses the same 50% of drops for its configuration phase. The remaining 50% of drops are then used for evaluation. This drop-based partitioning evaluates the ability of each method to generalize to previously unseen network topologies. Each drop corresponds to a distinct spatial realization of the MD-MIMO deployment, with different RU and UE locations and, consequently, different large-scale and small-scale channel characteristics. Therefore, evaluating on unseen drops introduces a mismatch between the training/configuration environment and the testing environment. This setting is representative of practical MD-MIMO systems, where it is difficult to collect an offline dataset that exhaustively covers all possible RU and UE placements, mobility patterns, blockage conditions, and propagation geometries. As a result, some degree of mismatch between the data used for training or configuration and the channels encountered during deployment should be expected. The drop-based dataset split is used to explicitly model this practical generalization challenge.

We observe that the performance gap between configured WESN and ChannelMamba is significantly larger under the spatial partitioning setting compared to the temporal partitioning setting. This indicates that generalizing to unseen spatial configurations (i.e., new RU and UE placements) is substantially more challenging than generalizing to unseen time instances within the same spatial topology. This observation is particularly important in the MD-MIMO setting. Due to the mobility of both RUs and UEs, the system experiences a continuously evolving set of spatial topologies, making it extremely difficult to collect sufficiently rich offline datasets that cover all possible configurations. As a result, offline-trained models such as ChannelMamba face inherent limitations in generalization. In contrast, online learning approaches can adapt to the current environment using limited real-time observations, making them better suited for MD-MIMO channel prediction under practical conditions. Within the on-line learning approaches, the configured WESN is shown to offer the best performance, while having lower computational complexity than full Kalman filtering.

VIII. CONCLUSIONS

In this paper, we investigated the impact of channel prediction on the performance of Mobile Distributed MIMO (MD-MIMO) systems under practical constraints, including codebook-based CSI feedback, feedback delay and mobility-induced channel aging. We showed that accurate channel prediction is a critical enabler for coherent joint transmission (CJT), particularly in high-mobility environments where feedback delay severely degrades precoder effectiveness. To address this challenge, we introduced a configured Windowed Echo State Network (WESN) framework that leverages the structure of steady-state Kalman predictors to design reservoir dynamics, while retaining the flexibility of data-driven

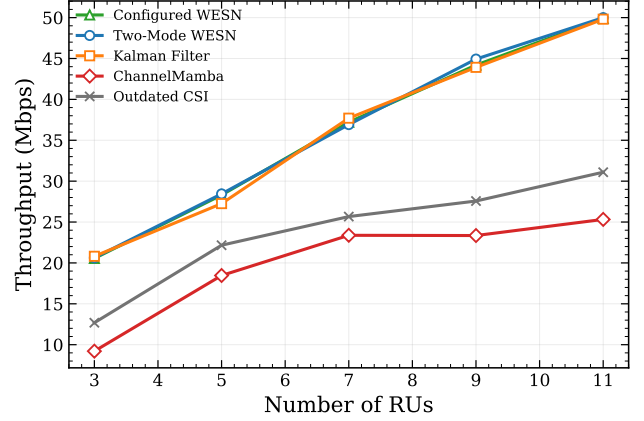


Fig. 9. Throughput vs number of RUs at 10 km/h under the drop-based split.

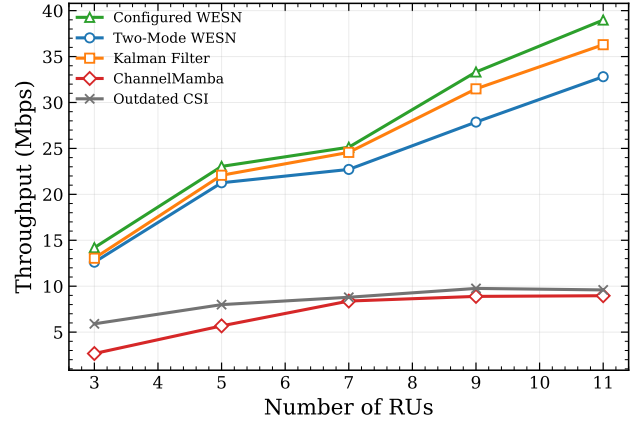


Fig. 10. Throughput vs number of RUs at 40 km/h under the drop-based split.

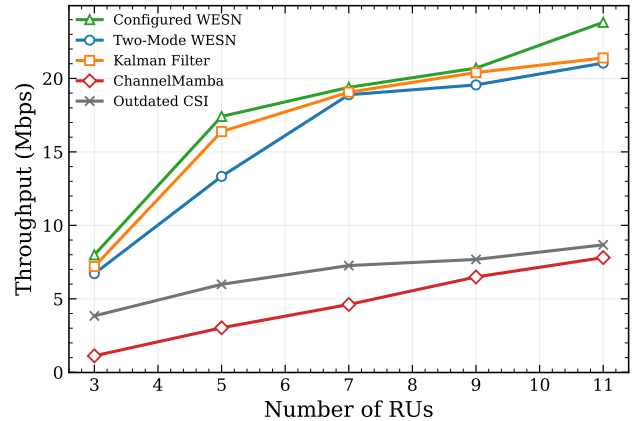


Fig. 11. Throughput vs number of RUs at 80 km/h under the drop-based split.

learning through low-complexity readout training. This hybrid approach combines the strengths of model-based and learning-based methods, enabling robustness to model mismatch and efficient online adaptation. Through extensive simulations, we demonstrated that the configured WESN consistently outperforms classical Kalman filtering as well as state-of-the-art learning-based baselines, especially in cases where imperfect knowledge of channel dynamics limits the effectiveness of model-based predictors.

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